
Questions from May 1, 2025

Question 1

In the proof of Theorem 4.5 why does the second equality in the next equation hold?

$$\mathbb{E}[\exp(itX_s(f))] = \exp\left(-\frac{1}{2}t^2\mathbb{E}[(X_s(f))^2]\right) = \exp\left(-\frac{1}{2}t^2\int_K f(-\Delta)^{-2s}f\,d\mu\right)$$

Question 2

In lemma 4.4 for $s = 0$ why is

$$(-\Delta_m)^0 f_m(p) = 1.$$

I cannot get it to fit with the definition of $(-\Delta_m)^{-s}$

Question 3

Again in the proof of lemma 4.4 you state that

$$\sup_m \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 < \infty.$$

My guess is that it is because V_m is a subset of the compact set K and $f_m \in \ell(V_m) \subset C(K)$.

Question 4

In the end of lemma 4.4, why does the following equality hold?

$$\frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \int_K f(x) P_t f(x) \, d\mu(x) \, dt = \int_K f(x) (-\Delta)^{-2s} f(x) \, d\mu(x)$$